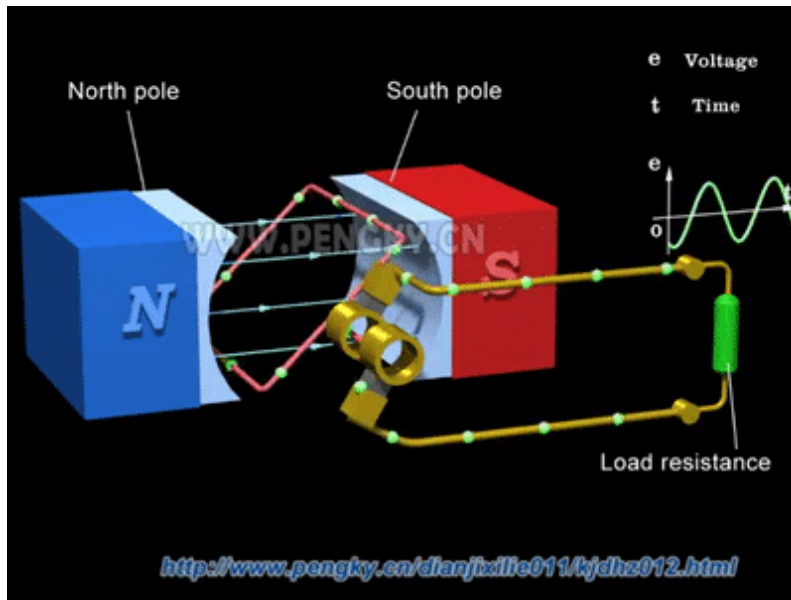


assignment 15 - mini assignment, Faradays law

 This is a preview of the draft version of the quiz

Started: Mar 23 at 6:40pm

Quiz Instructions



Question 1 17 pts



hint; find the emf first and $I = \text{emf}/R$

$\text{emf} = N \frac{d(BA)}{dt} = NA \frac{dB}{dt}$. Derive sine . remember B is in mT so divided by 1000.

note: 10π is your omega $\omega = 2\pi/T$. Use that for next question.

☐

d

☐

a

☐

b

☐

c

☐

e



Question 2 17 pts

If a magnetic field varies with time as given by $50 \sin(10\pi t)$ mT what is true

☐

The max B is 10mT and the frequency is 5Hz

☐

The max B is 50mT and the frequency is 5Hz

☐

The max value is 50mT and the frequency is 0.2 Hz

☐

The max B is 10mT and the frequency is 50Hz

☐

Question 3 17 pts



hint: we did such a problem in class. Remember angular frequency $\omega = 2\pi f$

☐

a

☐

b

☐

d

☐

c

☐

e

☐

Question 4 17 pts



hint: The slide 76 (faradays pdf) gives you the equation for the flux of the magnetic field poking through the area as a function of I . You have $a = 1\text{cm}$. $l = 1.5\text{m}$. $w = 5\text{cm}$ ($6\text{cm} - 1\text{cm}$). First write down the flux as a function of I . Then take the derivative to get the emf. the only thing that is changing is the current. So $\text{flux} = \dots dI/dt$. Use dI/dt which is given. The current is the $\text{emf}/\text{resistance}$.



b



e



c



a



d



Question 5 17 pts



Its easy. Remember slide 42 and how we derive the expression for the emf. $(\pi/2)$ is just the angle inside the sine. ($\omega = \text{angle/time}$). and 0.6 rad/s is just the ω .



a



c



b



e



d



Question 6 17 pts



we discussed in class how the induced current will oppose what gave rise to it. So the side a will be acted by a up force equals to the weight when terminal speed is reached. So magnetic force = weight. The force is IlB with I the induced current. The induced current is driven by the motional emf. See slide 29 and the work in class. The induced current is the $(\text{motional emf})/R$



e



c



a



d



b

Not saved

Submit Quiz